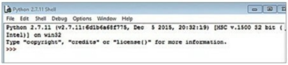

Fig. 11: Python IDLE (GUI) screen

Fig. 12: takabreak.py on Python GUI window

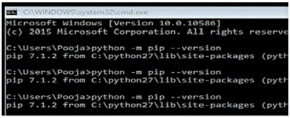

Fig. 13: Checking the pip version

Create new program

Run Python IDLE→File→New File.

Start writing the program in IDLE. The complete program has been provided in takabreak.py file (Fig. 12), and it is recommended to copy the code from this file, or use the file as it is, because Python is an indentation-sensitive language.

Importing modules

Let us start with importing different modules that would be required in the program.

import webbrowser
# importing webbrowser module to run videos from youtube

import random
# importing random module to select randomly from a list

import time
# importing time module for delays

import pyttsx
# text to speech module

The first three modules are packaged with Python standard library, so you do not have to install these. You can check all different modules in Python standard library from *https://docs.python.org/2/library/*

The fourth module (pyttsx) does not come packaged with Python standard library. So it needs to be installed before importing in the program. This module is used for making audio announcements.

Pyttsx installation

This project requires conversion of text to speech. To achieve this, a ready-made Python package called pyttsx is available. It provides the text-to-speech synthesis facility and is tested on Windows, Linux and Mac.

To install pyttsx package, you need a package manager like pip, which automatically downloads all necessary files. It also puts the files in correct directories to set up the system.

Python 2.7.11 already has this package included. To check if pip is installed, run the following command on Command Prompt (Fig. 13):

```
python -m pip –version
```

If pip is installed, the response will be something like:

```
pip 7.1.2 from C:\python27\lib\
site-packages (python 2.7)
```

Run the following command in Command Prompt to install pyttsx:

```
python -m pip install pyttsx
```

If you see the message to upgrade pip version, upgrade it by running the following command in Command Prompt:

```
Python -m pip install –upgrade pip
```

This completes the installation of pyttsx.

Installation of pywin32

pyttsx package is dependent on pywin32, so install it by downloading the same from *https://sourceforge.net/projects/pywin32/files/pywin32/*

The window shown in Fig. 14 will appear. Download the correct file depending on your Python version and its architecture installed on your PC. (Remember the Python version noted in previous steps.) In this case, it is the one highlighted in Fig. 14.

Once downloaded, install the file. Make sure that while installing, all applications using Python should be closed. This completes the installation of pywin32.

Functions used

1. pyttx setup functions:

```
engine = pyttsx.init()     # initialise
                 text to speech engine
engine.setProperty('rate', 150)   # set
                        speech rate
voices = engine.getProperty('voices')
                 # set voice property
engine.setProperty('voice', voices[0].
id)               # set female voice
```

Above functions initialise text-to-speech engine and set various parameters.

2. To play audio message:

```
engine.say('Hey SPARKY. Take a break
now.'
 ' Its been a long time you are sitting
on the chair.'
 ' I have selected some videos from the
internet to motivate you')
```

This is the audio message that will be played every time the software has to remind you to take a break.

3. Function used to play video from youtube:

```
webbrowser.open(random.choice(videos))
```

This will play random videos from the list in the program as given below:

```
videos = ('https://www.youtube.com/
watch?v=GSO6g3dNR7s',
```